

12.7 (a) The total momentum can be written as

$$\vec{p} = m\vec{v} - \frac{q}{c} \vec{A} + \vec{p}_{EM},$$

where $\vec{A}$ is the vector potential at (x_0, y_0, z_0) , and

$$\vec{p}_{EM} = \frac{1}{4\pi c} \int (\vec{E} \times \vec{B}) d^3x.$$

where $\vec{B} = -B\hat{z}$, and $\vec{E} = q(\vec{r} - \vec{r}_0)/|\vec{r} - \vec{r}_0|^3$. Since

$$\vec{E} \times \vec{B} = \frac{qB}{|\vec{r} - \vec{r}_0|^3} \left[-(y-y_0)\hat{i} + (x-x_0)\hat{j} \right],$$

$$\text{then } \vec{p}_{EM} = \frac{qB}{4\pi c} \int_0^a dx \int_{-b}^{+b} dy \int_{-b}^{+b} dz \frac{-(y-y_0)\hat{i} + (x-x_0)\hat{j}}{[(x-x_0)^2 + (y-y_0)^2 + (z-z_0)^2]^{3/2}}$$

$$= \hat{j} \frac{qB}{2c} \int_0^a dx \int_0^{+b} (x-x_0) \frac{p}{[(x-x_0)^2 + p^2]^{3/2}} dp$$

$$= \hat{j} \frac{qB}{2c} \int_0^a \frac{x-x_0}{|x-x_0|} dx = \hat{j} \frac{qB}{2c} \times \begin{cases} a, & x_0 < 0 \\ a-2x_0, & 0 \leq x_0 \leq a \\ -a, & x_0 > a \end{cases}$$

For the vector potential $\vec{A}$, we have some freedom in the choice of the gauge. Since Mr. Jackson wants some conservation of momentum, we will give him the conservation. To this end, let $\vec{A} = B(y, 0, 0)$ inside the strip. We can choose $\vec{A} \equiv 0$, outside the strip, so that the parallel component is continuous.

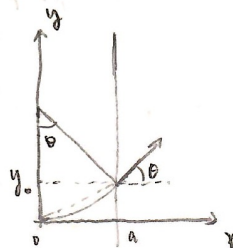
(b) Suppose the particle is incident along the x -axis. Then, at the incident location,

$x_0 = y_0 = z_0 = 0$, the momentum is

$$\vec{p}_i = m\vec{v} + \hat{j} \frac{qBa}{2c} = \left(mv, \frac{qBa}{2c}, 0 \right)$$

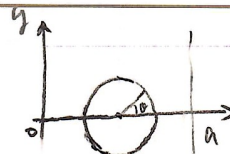
At the emerging location, $a = r \sin \theta$, where $r = cmv/qB$. Then

$$\begin{aligned} \vec{p}_f &= mv(\cos \theta, \sin \theta, 0) + \frac{qB}{2c} (0, -a, 0) \\ &= \left(mv \cos \theta, mv \sin \theta - \frac{qBa}{2c} \right) = \left(mv \cos \theta, \frac{qBa}{2c} \right) \end{aligned}$$



Therefore, the y -component of total momentum is conserved while x -component is not.

(c) When the particle is confined in the strip, assume the center of the circular motion is on the x -axis at x_0 . Then, the momentum at any time is



$$\vec{p} = mv(-\sin\theta, \cos\theta, 0) + \frac{qB}{c}(r\sin\theta, 0, 0) + \frac{qB}{2c}(0, a-2(x_0+r\cos\theta), 0)$$

With $r = \frac{cmv}{qB}$. Then,

$$\vec{p} = (0, mv\cos\theta + \frac{qB}{2c}(a-2x_0) - mv\cos\theta, 0)$$

$$= (0, \frac{qB}{2c}(a-2x_0), 0)$$

which indicates that the momentum is conserved

Interestingly, if we choose the symmetric gauge, $\vec{B} = \frac{B}{2}(y, -x, 0)$, neither component is conserved.

~~To have conservation of momentum, we need to consider a closed system, where all particles and fields are included. Here, we have ignored the charge and current for the creation of the magnetic field, which is treated as a static object and is not affected by the motion of the particles. Therefore, it is not surprising and does not break any physical law if the momentum is not conserved here.~~